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Foreword

This European Telecommunication Standard (ETS) has been produced by the Terrestrial Trunked Radio (TETRA) Project of the European Telecommunications Standards Institute (ETSI).

This ETS consists of 4 parts as follows:

Part 1: "Radio";

Part 2: "Protocol testing specification for Voice plus Data (V+D);

Part 4: "Protocol testing specification for Direct Mode Operation (DMO);

Part 5: "Security".

Transposition dates	
Date of adoption of this ETS:	4 June 1999
Date of latest announcement of this ETS (doa):	30 September 1999
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Date of withdrawal of any conflicting National Standard (dow):	31 March 2000

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1 Scope

This ETS contains the Test Specifications (i.e. Test Suite Structure and Test Purposes (TSS&TP) and the Abstract Test Suites (ATS)) to test conformity of products to the TETRA Direct Mode Operation (DMO) protocols. This ETS is divided into several parts, each one dealing with one TSS&TP or one ATS for the test of a layer 2 or layer 3 protocol for DMO. This present part 4-1 deals with TSS&TP for Mobile Station to Mobile Station (MS-MS) Air Interface (AI) protocol.

Testing of security features is outside the scope of this ETS.

The objective of this test specification is to provide a basis for approval tests for TETRA equipment giving a high probability of air interface inter-operability between different manufacturer's TETRA equipment.

The ISO standard for the methodology of conformance testing, ISO/IEC 9646-1 [4] and ISO/IEC 9646-2 [5], as well as the ETSI methodology for conformance testing, ETS 300 406 [3], are used as the basis for the test methodology.

2 References

This ETS incorporates by dated and undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

[1] ETS 300 396-3 (1997): "Terrestrial Trunked Radio (TETRA); Technical requirements for Direct Mode Operation (DMO); Part 3: Mobile Station to Mobile Station (MS-MS) Air Interface (AI) protocol".

[2] ETS 300 396-8-1 (1997): "Terrestrial Trunked Radio (TETRA); Technical requirements for Direct Mode Operation (DMO); Part 8: Protocol Implementation Conformance Statements (PICS) proforma specification; Sub-part 1: Mobile Station to Mobile Station (MS-MS) Air Interface (AI)".

[3] ETS 300 406 (1995): "Methods for Testing and Specification (MTS); Protocol and profile conformance testing specifications; Standardization methodology".

[4] ISO/IEC 9646-1 (1994): "Information technology; Open Systems Interconnection; Conformance testing methodology and framework; Part 1: General concepts". (See also ITU-T Recommendation X.290).

[5] ISO/IEC 9646-2 (1994): "Information technology; Open Systems Interconnection; Conformance testing methodology and framework; Part 2: Abstract Test Suite specification". (See also ITU-T Recommendation X.291).

3 Definitions and abbreviations

3.1 TETRA definitions

For the purposes of this ETS, the definitions given in ETS 300 396-3 [1] apply.

3.2 TETRA abbreviations

For the purposes of this ETS the following TETRA abbreviations apply:

CM	Circuit Mode
DMCC	Direct Mode Call Control
DMO	Direct Mode of Operation
FCS	Frame Check Sequence
MAC	Medium Access Control
MNI	Mobile Network Identity
MS	Mobile Station
NWK	Network. Layer 3 of the TETRA protocol stack
SDS	Short Data Services
SDU	Service Data Unit
TX	Transmit
RX	Receive

3.3 ISO 9646 definitions

For the purposes of this ETS the following ISO 9646-1 [4] terms and definitions apply:

Implementation Conformance Statement (ICS)
Implementation Under Test (IUT)
Implementation eXtra Information for Testing (IXIT)
Protocol Implementation Conformance Statement (PICS)
Protocol Implementation eXtra Information for Testing (PIXIT)

3.4 ISO 9646 abbreviations

For the purposes of this ETS the following ISO 9646-1 [4] abbreviations apply:

IUT	Implementation Under Test
PDU	Protocol Data Unit
PICS	Protocol Implementation Conformance Statement
PIXIT	Protocol Implementation eXtra Information for Testing
TP	Test Purpose
TSS	Test Suite Structure

4 Test Suite Structure (TSS)

4.1 NWK layer test groups

The first level separates the NWK layer (or layer 3) in different protocols (Circuit mode, Short Data Service). Next level splits protocol testing into functional test groups according to the type of testing: Capability test (CA), Valid Behaviour (BV) and Timer tests (TI). Further level classifies the possible operations in each protocol condition or state.

The following list defines the NWK layer test group names and identifiers used for those:

MS-MS Direct Mode Call Control (DMO_MSMS_DMCC)
- Circuit mode (CM);
  - Capability tests (CA);
  - Valid Behaviour tests (BV);
    - from Idle state(ID);
    - from Idle state, channel busy (IB);
    - from TX occupation State (TXO);
    - from RX occupation State (RO);
    - from TX Reservation (TR);
    - from RX Reservation State (RR);
  - Timer Tests (TI).

- Short Data Service (SDS);
  - Capability tests (CA);
  - Valid Behaviour tests (BV);
    - from Idle state(ID);
    - from Idle state, channel busy (IB);
    - from RX occupation State (RO);
    - from TX Reservation (TR);
    - from RX Reservation State (RR).
  - Timer Tests (TI).

4.2 MAC layer test groups

The first level of the MAC test groups separates the MAC test suite in functional test groups: CA, BV and TI. The second level of the test subgroups is used to form a division of protocol requirements.

The following list defines the MAC layer test group names and identifiers:

MS-MS MAC layer (DMO_DMO_MSMS_MACC)
- Capability tests (CA);
- Valid behaviour tests (BV);
  - Channel usage (CU);
  - Signalling messages (SM);
  - Traffic mode (TM).
- Timer tests (TI).

4.3 Test group description

Capability (CA) tests provide limited testing that the observable capabilities of the IUT are in accordance with the conformance requirements and the additional capabilities claimed in the PICS/PIXIT.

The Valid Behaviour (BV) group tests an IUT in response to valid behaviour of the test system. "Valid" means that a test event is syntactically and contextually correct. All test cases in the valid behaviour group are intended to verify as thoroughly as possible the various functions of the protocol.

Different timers are defined to supervise the various state transitions. The Timer (TI) test group is intended to verify that the IUT is reacting properly to an expiry of one of the timers or to a counter mismatch.

5 Introduction to Test Purposes (TPs)

The TPs for each test suite are defined in clause 6 of this ETS for NWK layer and MAC layer.

5.1 Test purpose definition conventions

5.1.1 TPs descriptions

Each TP is described using text presented in a table.

The table contains the following information:

Table 0

TP-Name The TP name is a unique identifier, specified according to the TP naming conventions defined in the subclause 5.1.2. (it is also the name of the corresponding test case)	**Reference:** reference to the paragraph number of specification ETS 300 396-3 [1] stating this conformance requirement. For example: ETS 300 396-3 [1], 6.2.5.1
Purpose	purpose of the test itself, indicating for example the test performed against a requirement of the protocol, described by this test purpose. Example: test of changeover initiated from RX reservation state
Test description	body of the test
Pass criteria	visible action to be observed at PCO to declare that the IUT passes the test and conforms to the specifications
Selection	expression based on ETS 300 396-8-1 [2] PICS statements, used to select or deselect the corresponding test case according to the options of the implementation.
Preamble	"None" or name of the preamble procedure bringing the IUT from idle state to the state required to run the test. For example: idle_to_RX_reservation
Postamble	"None" or name of the postamble to bring the IUT back to idle state, for example: RX_occupation_to_idle.

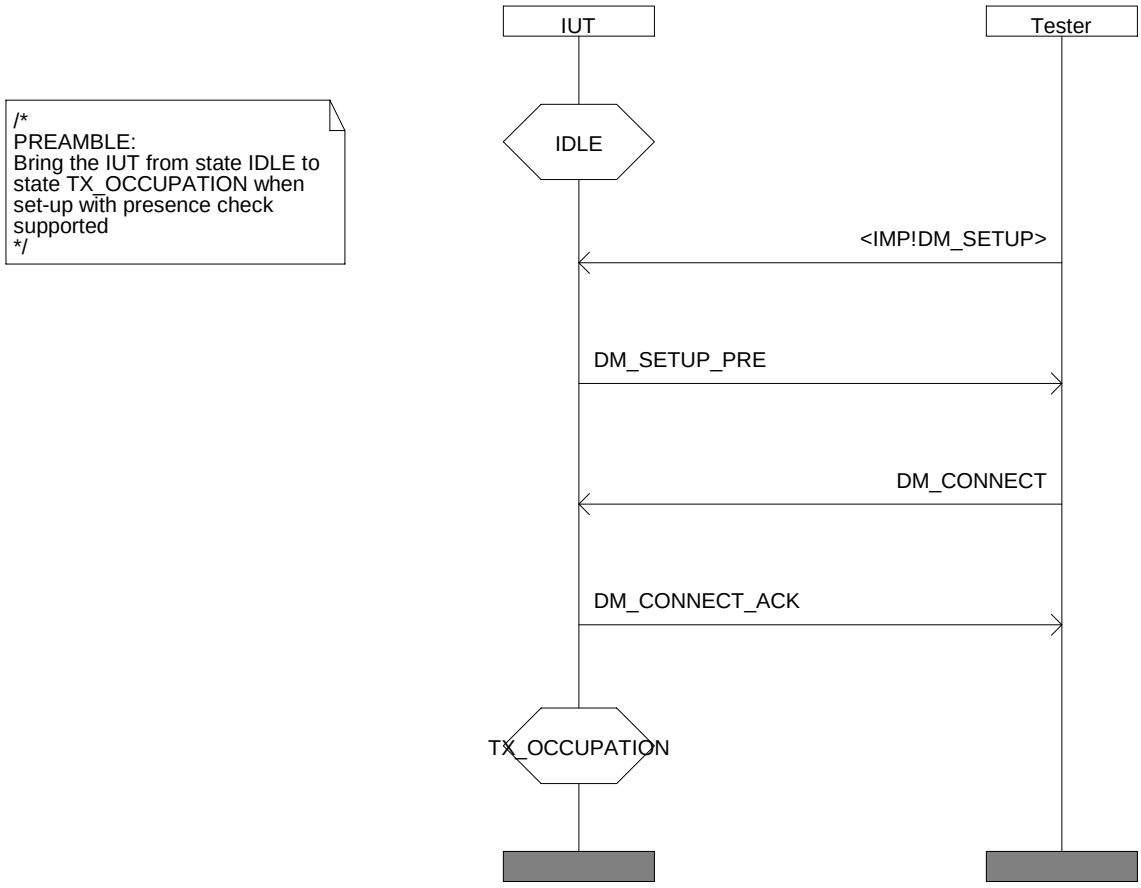
The preambles and postambles are described using MSCs and are shown in the following subclauses.

5.1.2 Preamble descriptions

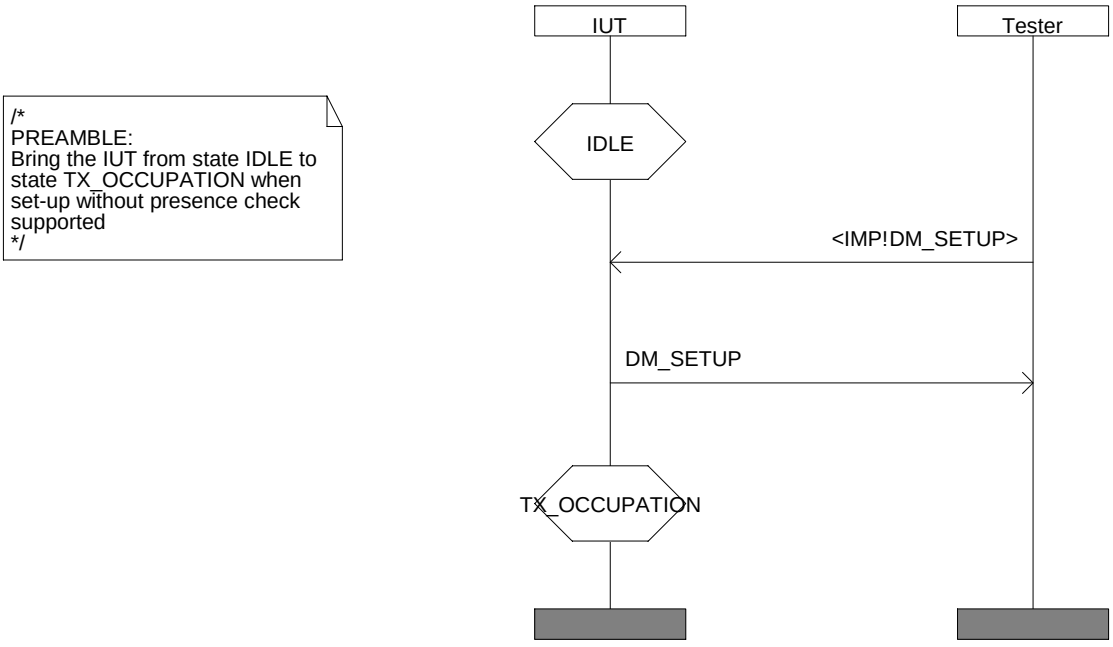
Preambles are used to bring the IUT from the idle state to the state where the test takes place. As the protocol has different options, as for instance the use of presence check or the absence of presence check, there are several ways to reach a given state. The preamble has to be chosen according to the IUT capabilities and the implemented options.

5.1.2.1 Preamble idle_to_TX_occupation: From Idle state to Call Active TX Occupation

With presence check

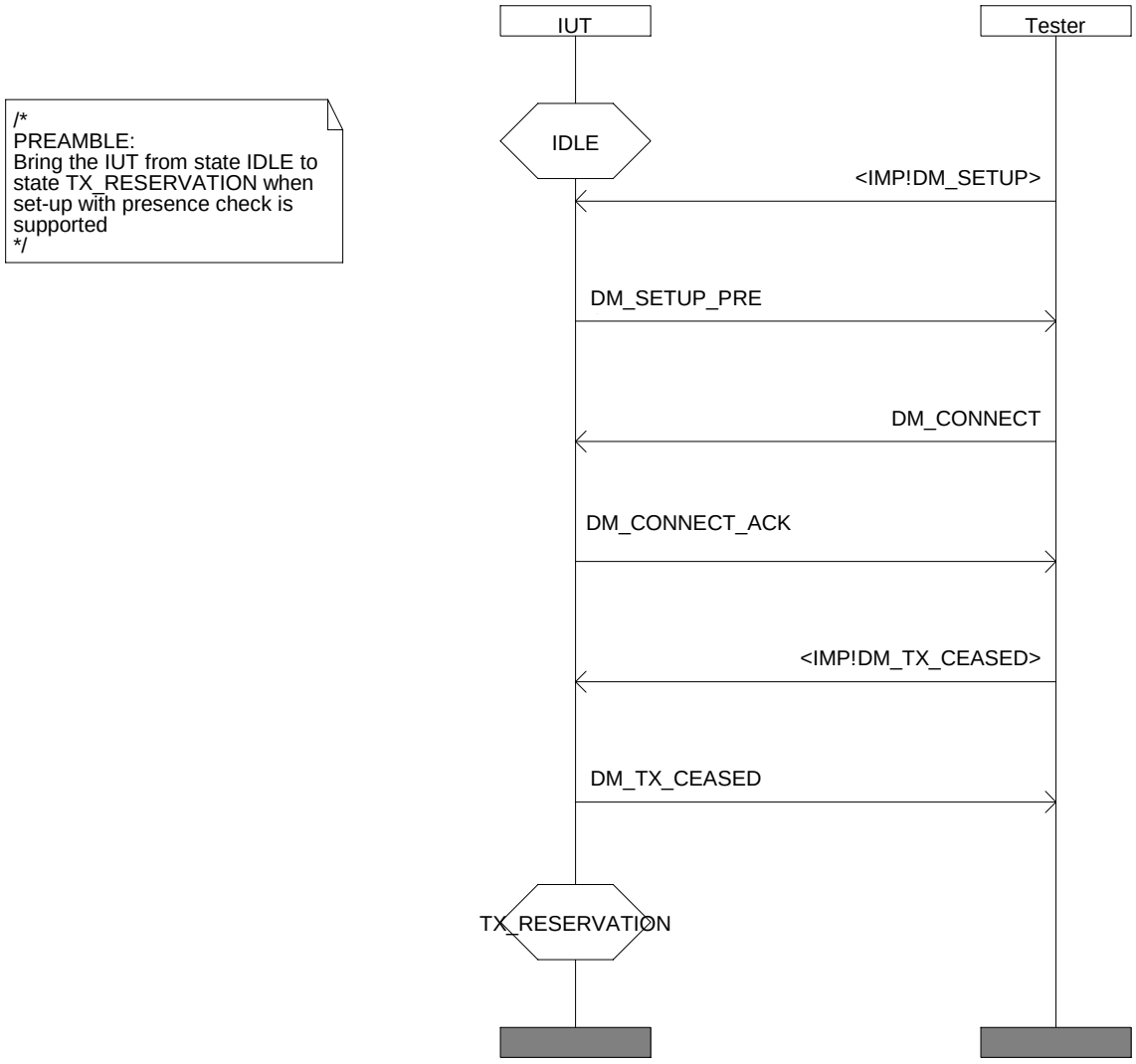


Without presence check

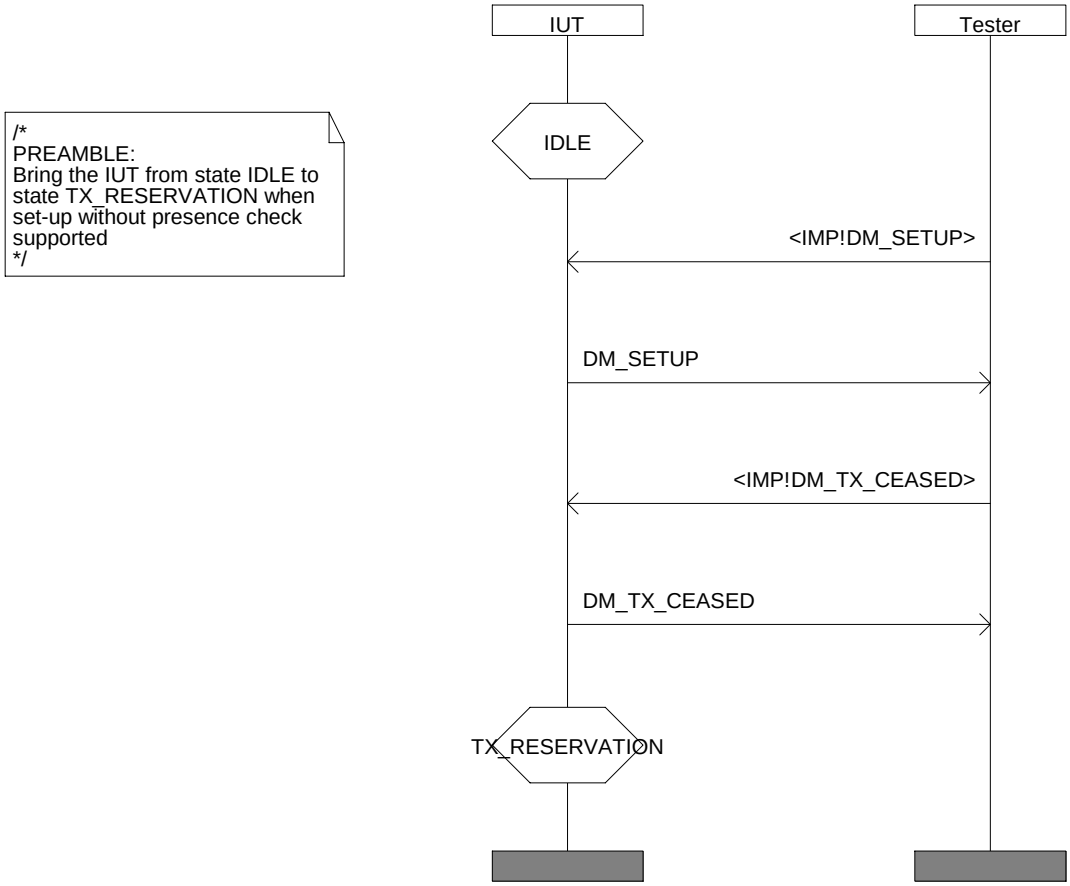


5.1.2.2 Preamble idle_to_TX_reservation: From Idle state to Call Active TX Reservation

With presence check

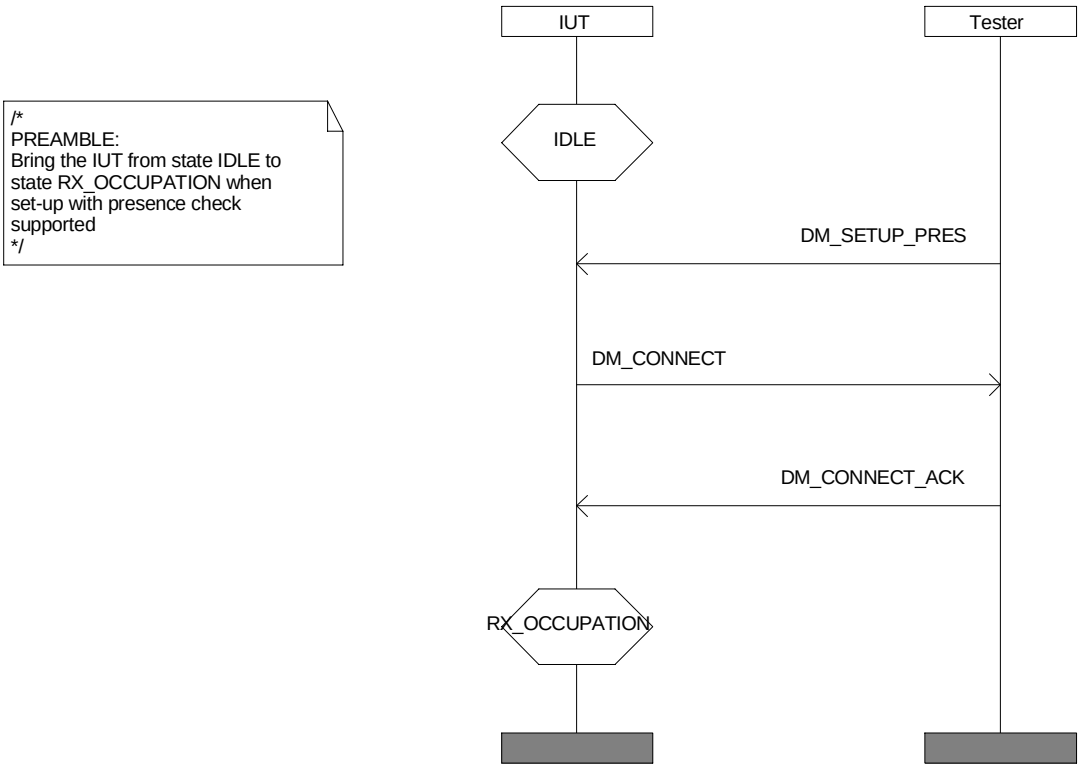


Without presence check

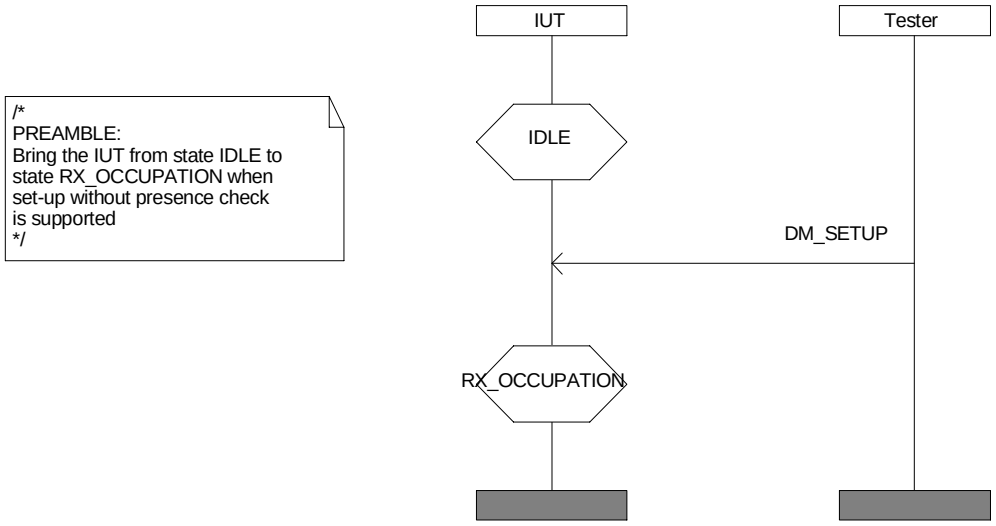


5.1.2.3 Preamble idle_to_RX_occupation: From Idle state to Call Active RX Occupation

With presence check

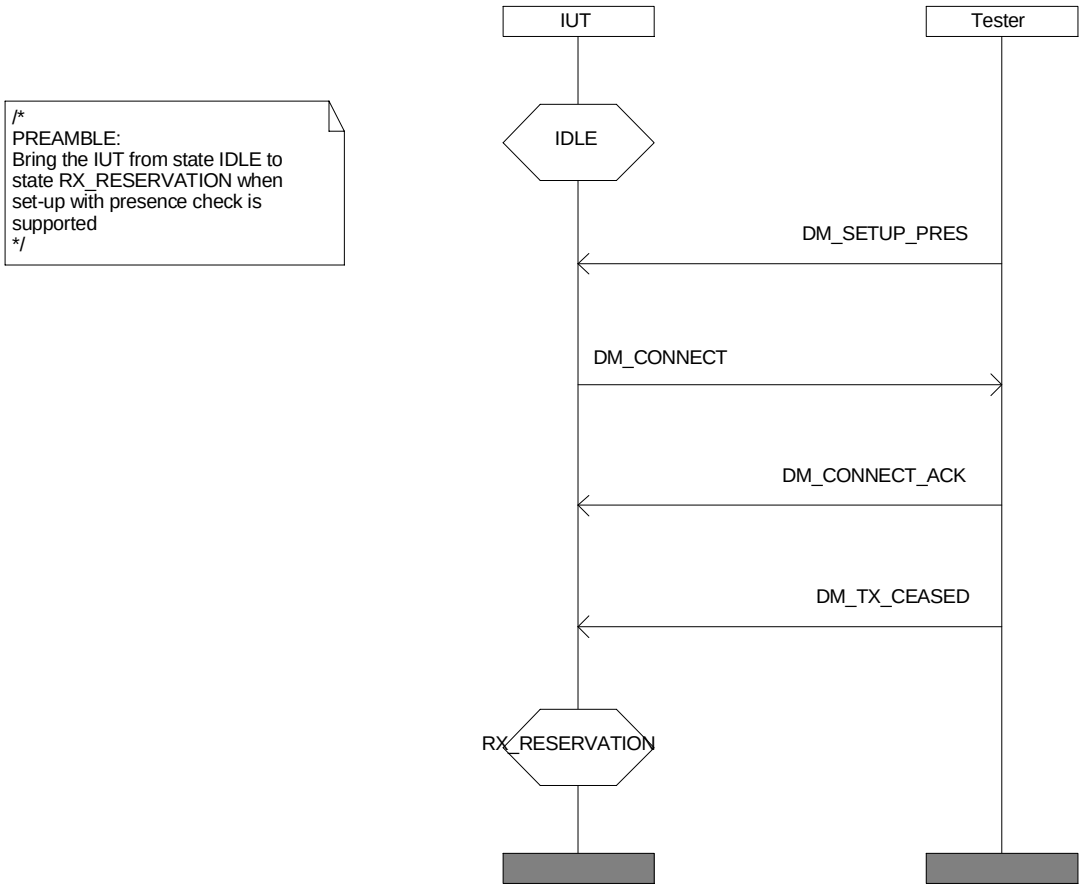


Without presence check

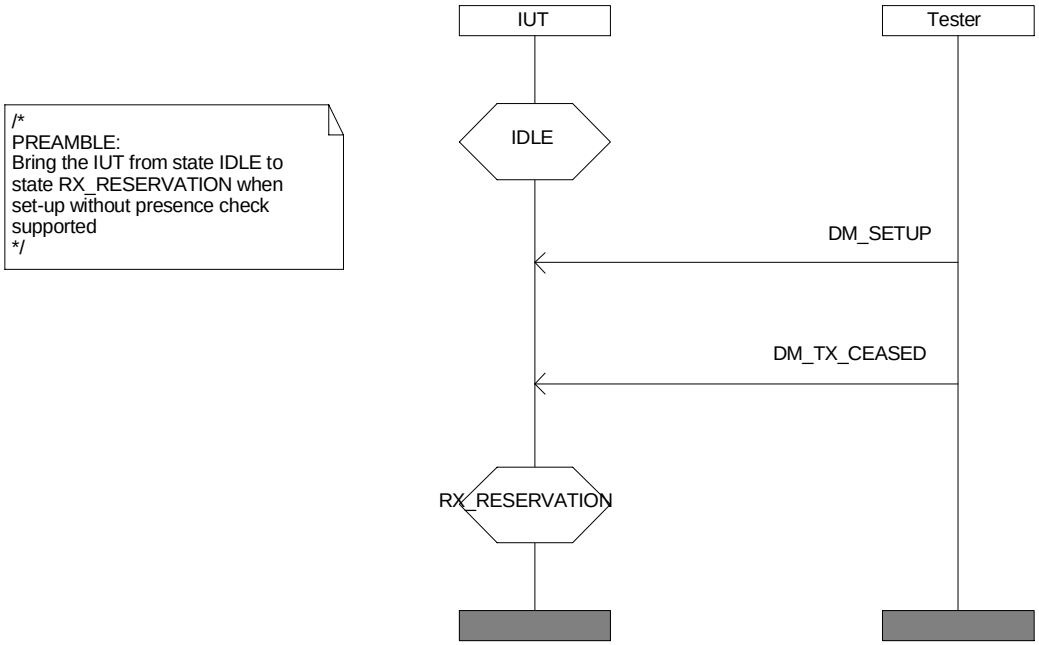


5.1.2.4 Preamble idle_to_RX_reservation

With presence check

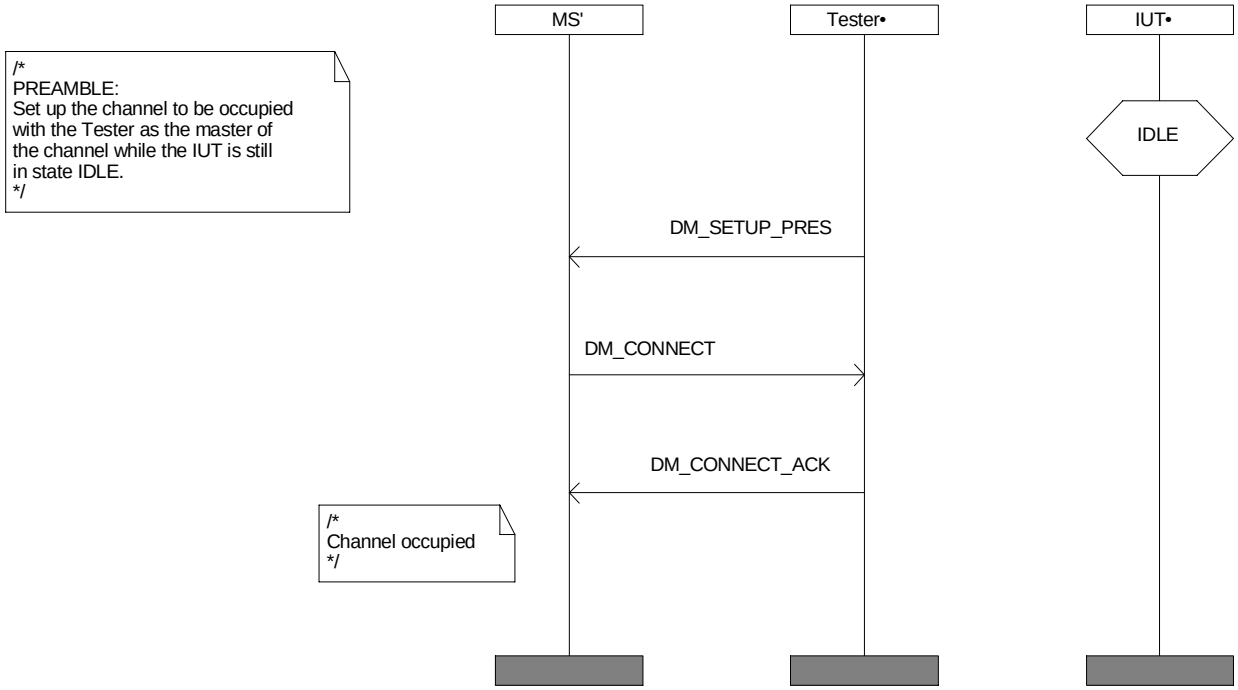


Without presence check



5.1.2.5 Preamble idle_channel_occupation

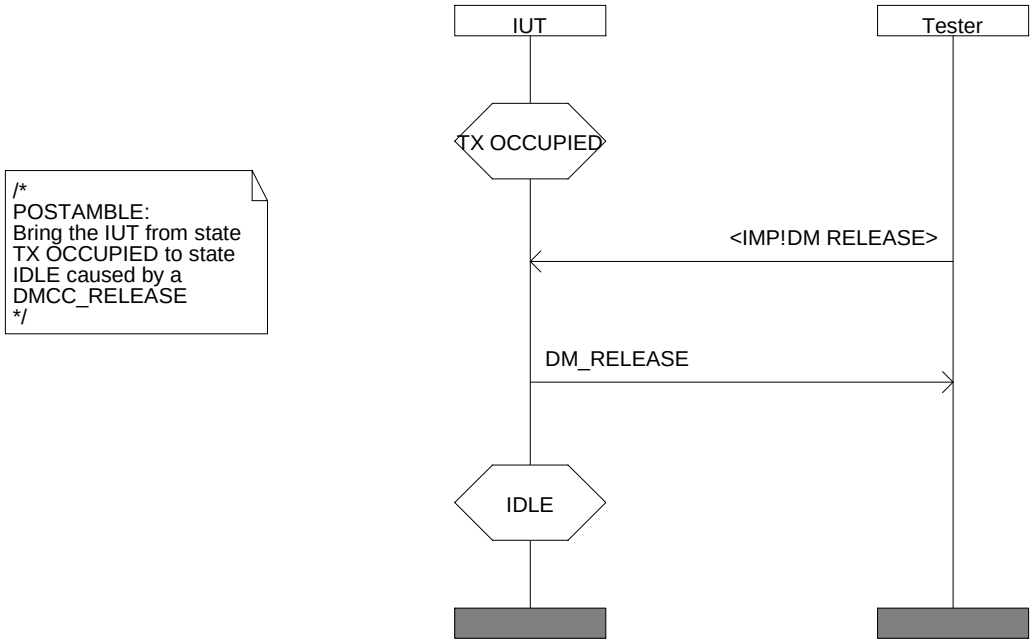
Without presence check



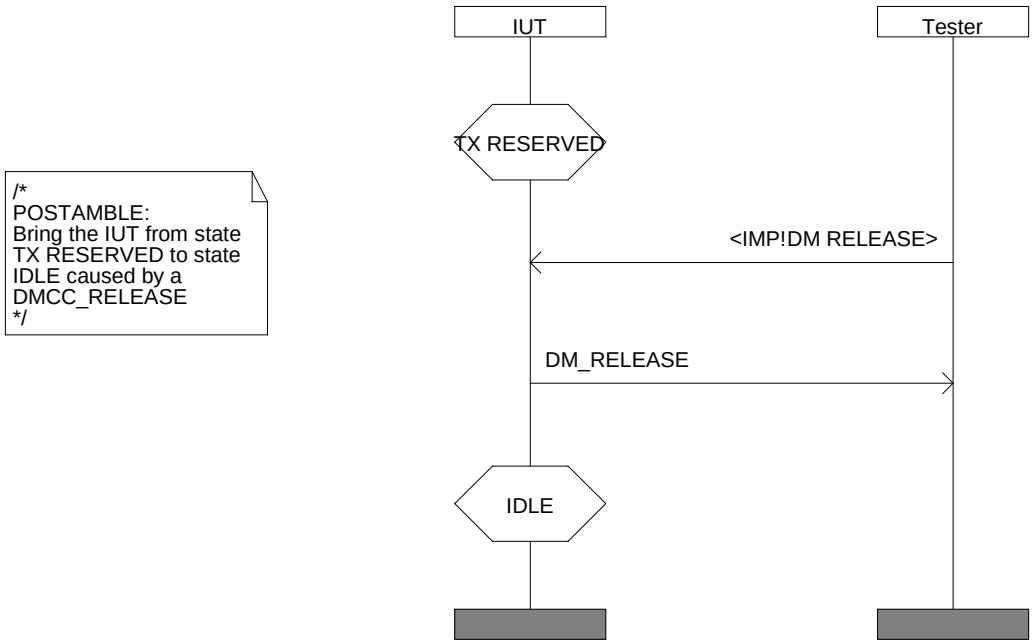
5.1.3 Postamble descriptions

Postambles are used to bring the IUT from the state ending the test, to the idle state.

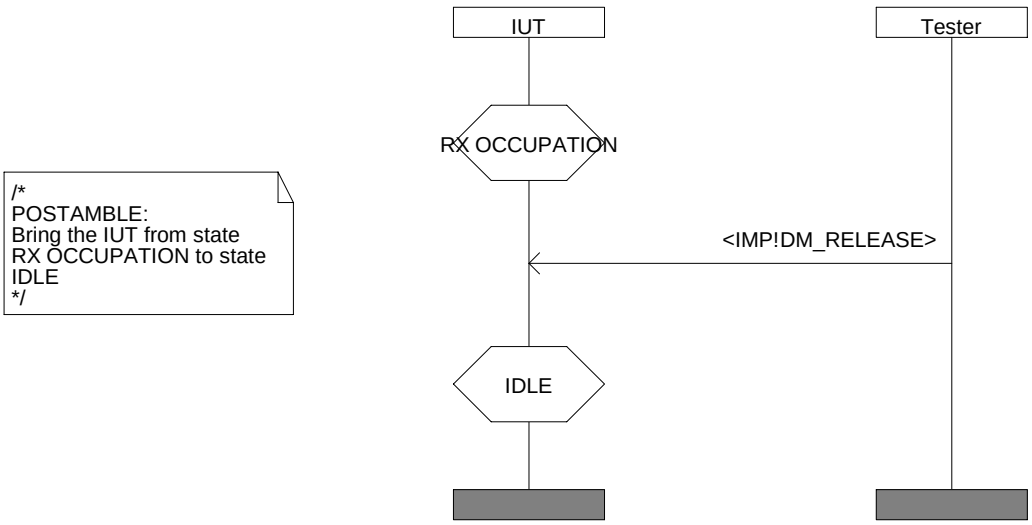
5.1.3.1 Postamble TX_occupation_to_idle: From Call Active TX Occupation state to Idle



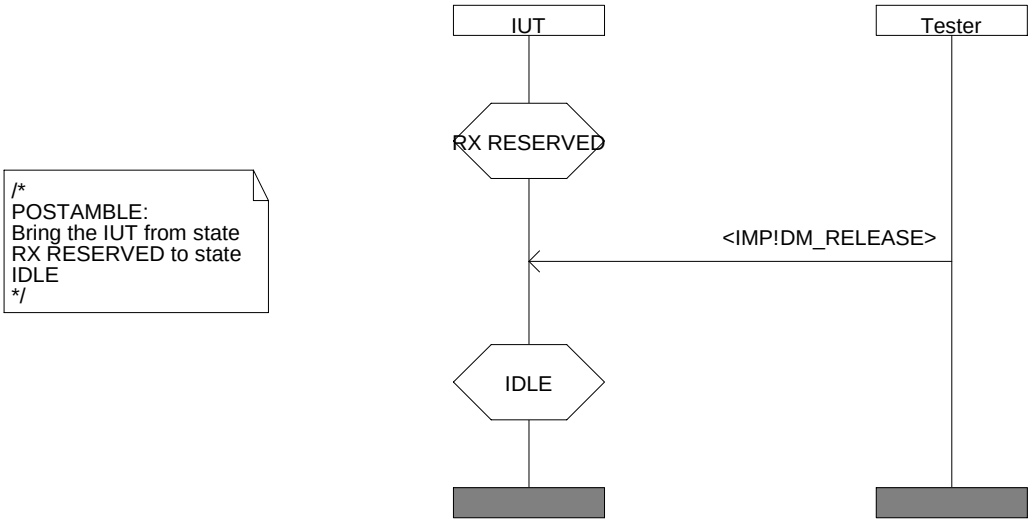
5.1.3.2 Postamble TX_reservation_to_idle: From Call Active TX Reserved state to Idle



5.1.3.3 Postamble RX_occupation_to_idle: From Call Active RX Occupation state to Idle



5.1.3.4 Postamble RX_reservation_to_idle: From Call Active RX Reserved state to Idle



5.2 Test purpose naming conventions

The identifier of the test purpose is built according to table 1:

Table 1: Test purpose naming convention

DMO/<ts>/<fm>/<ss>/<tt>/<tsg>/<nn>		
<ts> = test suite type	MSMS	MS to MS
	REPx	Repeater type x (see 300 394-4-4)
	GATE	Gateway (see 300 394-4-7)
<fm> = functional module or subentity (layer 3 only)	DMCC	Direct Mode Call Control (layer 3)
	MAC	Upper MAC (layer 2)
<ss> = test group	letters such as:	abbreviation of the group name (optional)
	CM	Circuit Mode (layer 3)
	SDS	Short Data Service (layer 3)
tt = Type of testing	CA	Capability Tests
	BV	Valid Behaviour Tests
	BI	Invalid Behaviour Tests
	TI	Timer expiry and counter mismatch tests
<tsg> = test subgroup	two letters	subgroup name (optional)
<nn> = sequential number	01-99	Test Purpose Number

5.3 Selection expressions

A test case, based on a test purpose described here, can be selected or deselected from the test suite, according to the evaluation of selection expressions which reflect the capabilities supported or not by the implementation under test.

It appears that some selection expressions are quite complex, mainly the ones used for the MAC layer test suite, as they are based on a rather long combinaison of PICS statements.

To ease the readability of the test purposes, these complex selection expressions are replaced by generic names which are defined here, and which represent by definition the selection expression themselves.

The following table defines the generic names together with the conditions associated with each one.

Table 2

Selection expression identifier	Selection expression (References to ETS 300 396-8-1)	Static capabilities associated with this selection
Initiate_CM_call	A.3/1 OR A.4/1 OR A.4/2	Initiate group CM call or Initiate individual CM call without presence check or Initiate individual CM call with presence check
Initiate_SDS_call	A.8/1 OR A.9/1 OR A.9/2 OR A.9/3	Send group unacknowledged SDS or Send individual unacknowledged SDS or Send acknowledged SDS or Sending acknowledged SDS with data in ACK
Initiate_CM_or_SDS_call	A.3/1 OR A.4/1 OR A.4/2 OR A.8/1 OR A.9/1 OR A.9/2 OR A.9/3	Initiate group CM call or Initiate individual CM call without presence check or Initiate individual CM call with presence check or Send group unacknowledged SDS or Send individual unacknowledged SDS or Send acknowledged SDS or Send acknowledged SDS with data in ACK
Receive_Ackd_CM_or_SDS_call	A.2/6 OR A.11/2 OR A.11/3	Accept CM call setup with presence check, Receive acknowledged SDS, Receive acknowledged SDS with data in ACK

6 DMO MS-MS test purposes

6.1 DMCC Circuit Mode (CM) tests

Test group objective: To test the behaviour of the DMCC CM entity of the IUT.

Condition: IUT implements the CM.

6.1.1 MS-MS CM capability tests

To test the basic capabilities of the CM module of the IUT, when operating in group address mode (without presence check) or in individual address mode (with or without presence check).

DMO_MSMS_DMCC_CM_CA_01	**Reference:** ETS 300 396-3 [1], 6.2.1.1, 6.2.4.1
Purpose	Setup and terminate a group call without presence check
Test description	The tester sends an implicit send to the IUT to cause a call setup.
Pass criteria 1	The IUT sends DM-SETUP to the tester
Test description	The tester sends an implicit send to the IUT to terminate the call
Pass criteria 2	The IUT sends DM-TX CEASED to the tester
Selection ETS 300 396-8-1 [2]	A.3/1 Setup procedure, group call address
Preamble	None
Postamble	TX_reservation_to_idle

DMO_MSMS_DMCC_CM_CA_02	**Reference:** ETS 300 396-3 [1], 6.2.2.1, 6.2.4.1
Purpose	Setup and terminate an individual call with presence check
Test description	The tester sends an implicit send to the IUT to cause a call setup. The IUT sends DM-SETUP PRES to the tester. The tester sends DM-CONNECT to the IUT
Pass criteria 1	The IUT sends DM-CONNECT ACK to the tester
Test description	The tester sends an implicit send to the IUT to terminate the call
Pass criteria 2	The IUT sends DM-TX CEASED to the tester
Selection ETS 300 396-8-1 [2]	A.4/2 Setup individual call with presence check
Preamble	None
Postamble	TX_reservation_to_idle

DMO_MSMS_DMCC_CM_CA_03	**Reference:** ETS 300 396-3 [1], 6.2.1.1, 6.2.4.1
Purpose	Establish and terminate an individual call, when operating without presence check
Test description	The tester sends an implicit send to the IUT to cause a call setup.
Pass criteria 1	The IUT sends DM-SETUP to the tester
Test description	The tester sends an implicit send to the IUT to terminate the call
Pass criteria 2	The IUT sends DM-TX CEASED to the tester
Selection ETS 300 396-8-1 [2]	A.4/1 Setup individual call without presence check
Preamble	None
Postamble	TX_reservation_to_idle

6.1.2 MS-MS CM valid behaviour tests

6.1.2.1 The IUT is in idle state, DMO channel is free

DMO_MSMS_DMCC_CM_BV_ID_01	**Reference:** ETS 300 396-3 [1], 6.2.2.1
Purpose	Establish an outgoing call with presence check initiated from idle state and DMO channel free.
Test description	The tester sends an implicit send to the IUT to cause a call setup. Then the IUT sends DM-SETUP PRES received by the tester, which sends back DM-CONNECT
Pass criteria	The IUT sends DM-CONNECT ACK PDU to the tester
Selection ETS 300 396-8-1 [2]	A.4/2 Setup individual call with presence check
Preamble	None
Postamble	TX_occupation_to_idle

DMO_MSMS_DMCC_CM_BV_ID_02	**Reference:** ETS 300 396-3 [1], 6.2.1.2
Purpose	Receive an incoming call without presence check
Test description	The tester sends DM-SETUP PDU to the IUT.
Pass criteria	To check that IUT reaches "call_active_RX_occupation" state, the tester sends DM-TX CEASED which brings the IUT to "call_active_RX_reservation". During the reservation period, when the IUT attempts a call setup, it shall issue a DM-TX REQUEST to initiate a changeover, and this is the pass criteria. Note: This call setup is controlled by the tester using an implicit send containing a "DMCC_SETUP_request".
Selection ETS 300 396-8-1 [2]	A.1/1 Circuit mode call
Preamble	None
Postamble	Tester issues a DM-REJECT followed by RX_Reservation_to_idle

DMO_MSMS_DMCC_CM_BV_ID_03	**Reference:** ETS 300 396-3 [1], 6.2.2.2
Purpose	Receive an incoming call with presence check
Test description	The tester sends DM-SETUP PRES to the IUT which sends back DM-CONNECT. The tester responds with DM-CONNECT ACK
Pass criteria	To check that IUT reaches state "call_active_RX_occupation" when receiving DM-CONNECT ACK, the tester sends DM-TX CEASED which brings the IUT to "call_active_RX_reservation". During the reservation period, when the IUT attempts a call setup, it shall issue a DM-TX REQUEST to initiate a changeover, and this is the pass criteria. Note: This call setup is controlled by the tester using an implicit send containing a "DMCC_SETUP_request".
Selection ETS 300 396-8-1 [2]	A.2/6 Accept call setup with presence check
Preamble	None
Postamble	Tester issues a DM-REJECT followed by RX_Reservation_to_idle

DMO_MSMS_DMCC_CM_BV_ID_04	**Reference:** ETS 300 396-3 [1], 6.2.2.1
Purpose	Release a call setup attempt when receiving a disconnect
Test description	The tester sends an implicit send to the IUT to cause a call setup. Then the IUT sends DM-SETUP PRES to the tester. The tester sends DM-DISCONNECT to the IUT to reject the call
Pass criteria	The IUT sends DM-RELEASE to the tester and returns to idle
Selection ETS 300 396-8-1 [2]	A.4/2 Setup individual call with presence check
Preamble	None
Postamble	None

DMO_MSMS_DMCC_CM_BV_ID_05	**Reference:** ETS 300 396-3 [1], 6.2.2.1
Purpose	Release a call setup attempt when the offered Quality of Service is not acceptable to the DMCC
Test description	The tester sends an implicit send to the IUT to cause a call setup. Then the IUT sends DM-SETUP PRES to the tester. The tester sends DM-CONNECT to the IUT with an unacceptable QOS
Pass criteria	The QOS being not acceptable, the IUT sends DM-RELEASE to the tester and returns to idle
Selection ETS 300 396-8-1 [2]	A.4/2 Setup individual call with presence check
Preamble	None
Postamble	None

DMO_MSMS_DMCC_CM_BV_ID_06	**Reference:** ETS 300 396-3 [1], 6.2.1.1, 6.2.4.1, 8.5.7.2.1
Purpose	Pre-emption flags in DM-SETUP and DM-TX-CEASED PDU.
Test description	The tester sends an implicit send to cause a call setup.
Pass criteria 1	Verify that IUT sends the DM-SETUP PDU with the pre-emption flag set to 1.
Test description	The tester sends an implicit send to cause the Iut to terminate the call.
Pass criteria 2	Verify that the IUT sends the DM-TX CEASED PDU with the request and changeover flags set to 1.
Selection ETS 300 396-8-1 [2]	A.4/2 Setup individual call with presence check or OR A.4/1 Setup individual call without presence check
Preamble	None
Postamble	None

6.1.2.2 IUT is in idle state, DMO channel is busy

DMO_MSMS_DMCC_CM_BV_IB_01	**Reference:** ETS 300 396-3 [1], 6.2.6
Purpose	Initiate call pre-emption, to establish a new CM call, from an MS not involved in the current call
Test description	The tester sends an implicit send to the IUT to cause a call setup. As the channel is busy, the IUT initially sends a DM-PREEMPT to the tester, which responds by sending a DM-PREEMPT_ACCEPT.
Pass criteria	The IUT sends DM-SETUP or DM-SETUP PRES to the tester according to the IUT capability
Selection ETS 300 396-8-1 [2]	A2/13 Initiating a new call by pre-emption
Preamble	idle_channel_occupation
Postamble	None (after waiting time over T303 and N303 times)

6.1.2.3 IUT is in TX occupation state

DMO_MSMS_DMCC_CM_BV_TXO_01	**Reference:** ETS 300 396-3 [1], 6.2.4.1
Purpose	Initiate the release of a call
Test description	The tester issues an implicit send containing a "DMCC_RELEASE_request" to the IUT.
Pass criteria	The IUT sends DM-RELEASE to the tester and returns to idle, state observable by the channel being free.
Selection ETS 300 396-8-1 [2]	Initiate_CM_call
Preamble	idle_to_TX_occupation
Postamble	None

DMO_MSMS_DMCC_CM_BV_TXO_02	**Reference:** ETS 300 396-3 [1], 6.2.4.1
Purpose	Initiate end of transmission (TX-ceased)
Test description	The tester issues an implicit send containing a "DMCC_TX_CEASED_request" to the IUT.
Pass criteria	The IUT sends TX CEASED to the tester and moves to state TX reservation.
Selection ETS 300 396-8-1 [2]	Initiate_CM_Call
Preamble	idle_to_TX_occupation
Postamble	TX_reservation_to_idle

DMO_MSMS_DMCC_CM_BV_TXO_03	**Reference:** ETS 300 396-3 [1], 6.2.4.1
Purpose	Receive pre-emption for an ongoing individual call
Test description	The tester sends a DM-PREEMPT to the IUT, containing the address of master
Pass criteria	The IUT sends back DM-PRE_ACCEPT and moves to "call_active_RX_reservation". During the reservation period, when the IUT attempts a call setup, it shall issue a DM-TX REQUEST to initiate a changeover, and this is the pass criteria. Note: This call setup is controlled by the tester using an implicit send containing a "DMCC_SETUP_request".
Selection ETS 300 396-8-1 [2]	Initiate_CM_Call
Preamble	idle_to_TX_occupation
Postamble	RX_Reservation_to_idle

DMO_MSMS_DMCC_CM_BV_TXO_04	**Reference:** ETS 300 396-3 [1], 6.2.4.1
Purpose	Receive pre-emption for a new individual call
Test description	The tester sends a DM-PREEMPT to the IUT, containing the address of a new pre-emptier
Pass criteria	The IUT sends back DM-PRE_ACCEPT to the pre-empter, followed by a DM-RELEASE to the slave and moves to idle (observable by the channel being free)
Selection ETS 300 396-8-1 [2]	Initiate_CM_Call
Preamble	idle_to_TX_occupation
Postamble	None

DMO_MSMS_DMCC_CM_BV_TXO_05	**Reference:** ETS 300 396-3 [1], 6.2.4.1
Purpose	Receive and reject pre-emption for a new individual call
Test description	The tester sends a DM-PREEMPT to the IUT, containing an unacceptable priority
Pass criteria	The IUT sends back DM-REJECT to the pre-empter
Selection ETS 300 396-8-1 [2]	Initiate_CM_Call
Preamble	idle_to_TX_occupation
Postamble	TX_occupation_to_idle.

6.1.2.4 IUT is in RX occupation state

DMO_MSMS_DMCC_CM_BV_RO_01	**Reference:** ETS 300 396-3 [1], 6.2.4.2
Purpose	Receive normal end of transmission (TX Cease)
Test description	The tester sends DM-TX CEASED to the IUT
Pass criteria	The IUT moves to state "call active RX Reservation". During the reservation period, when the IUT attempts a call setup, it shall issue a DM-TX REQUEST to initiate a changeover, and this is the pass criteria.
Selection ETS 300 396-8-1 [2]	A.1/1 Circuit mode call
Preamble	idle_to_RX_occupation
Postamble	RX_Reservation_to_idle

DMO_MSMS_DMCC_CM_BV_RO_02	**Reference:** ETS 300 396-3 [1], 6.2.4.2
Purpose	Initiate pre-emption to establish a call (either ongoing or new call)
Test description	The tester issues an implicit send containing a "DMCC_SETUP_request" to the IUT. The IUT sends DM-PREEMPT (address = master) to the tester, which accepts it by answering DM-PRE_ACCEPT
Pass criteria	The IUT sends DM-SETUP or DM-SETUP PRES to the tester according to the IUT capability
Selection ETS 300 396-8-1 [2]	A.2/12 Initiate pre-emption in ongoing call
Preamble	idle_to_RX_occupation
Postamble	In order to clear the call with presence check, the tester sends a DM-DISCONNECT PDU and waits for the DM-RELEASE PDU. In order to clear the call without presence check, the postamble TX_occupation_to_idle is used.

DMO_MSMS_DMCC_CM_BV_RO_03	**Reference:** ETS 300 396-3 [1], 6.2.4.2
Purpose	Handle the reject of a pre-emption
Test description	The tester issues an implicit send containing a "DMCC_SETUP_request" to the IUT. The IUT sends DM-PREEMPT (address = master) to the tester, which does not accept it and answers DM-REJECT
Pass criteria	The IUT stays in state "call_active_RX_occupation" when receiving DM-REJECT. To check it, the tester sends DM-TX CEASED which brings the IUT to "call_active_RX_reservation". During the reservation period, when the IUT attempts a call setup, it shall issue a DM-TX REQUEST to initiate a changeover, and this is the pass criteria. Note: This call setup is controlled by the tester using an implicit send containing a "DMCC_SETUP_request".
Selection ETS 300 396-8-1 [2]	A.2/12 Initiate pre-emption in ongoing call
Preamble	idle_to_RX_occupation
Postamble	Tester issues a DM-REJECT followed by RX_Occupation_to_idle

DMO_MSMS_DMCC_CM_BV_RO_04	**Reference:** ETS 300 396-3 [1], 6.2.4.2
Purpose	Reception of the ongoing call setup
Test description	The tester sends a DM-SETUP PRES PDU related to the ongoing call.
Pass criteria	Verify that the IUT sends the DM-CONNECT PDU.
Selection ETS 300 396-8-1 [2]	A.1/6 Accept call set-up with presence check
Preamble	idle_to_RX_occupation
Postamble	Tester sends the DM-CONNECT ACK PDU and then the postamble TX_occupation_to_idle is used to clear the call.

6.1.2.5 IUT is in TX reservation state

DMO_MSMS_DMCC_CM_BV_TR_01	**Reference:** ETS 300 396-3 [1], 6.2.5.1
Purpose	Initiate release of a call
Test description	The tester issues an implicit send containing a "DMCC_RELEASE_request" to the IUT.
Pass criteria	The IUT sends DM-RELEASE to the tester
Selection ETS 300 396-8-1 [2]	Initiate_CM_Call
Preamble	idle_to_TX_reservation
Postamble	None

DMO_MSMS_DMCC_CM_BV_TR_02	**Reference:** ETS 300 396-3 [1], 6.2.5.1
Purpose	Receive and accept pre-emption for a new call
Test description	The tester sends DM-PREEMPT to the IUT for a new call
Pass criteria	The IUT sends DM-PRE_ACCEPT to the tester
Selection ETS 300 396-8-1 [2]	Initiate_CM_Call
Preamble	idle_to_TX_reservation
Postamble	None

DMO_MSMS_DMCC_CM_BV_TR_03	**Reference:** ETS 300 396-3 [1], 6.2.5.1
Purpose	Receive and accept pre-emption for continuation of ongoing call
Test description	The tester sends DM-PREEMPT to the IUT for a call continuation
Pass criteria	The IUT sends DM-PRE_ACCEPT to the tester
Selection ETS 300 396-8-1 [2]	Initiate_CM_Call
Preamble	idle_to_TX_reservation
Postamble	RX_reservation_to_idle

DMO_MSMS_DMCC_CM_BV_TR_04	**Reference:** ETS 300 396-3 [1], 6.2.5.1
Purpose	Receive and accept changeover
Test description	The tester sends DM-TX REQUEST to the IUT indicating call continuation
Pass criteria	The IUT sends DM-TX ACCEPT to the tester
Selection ETS 300 396-8-1 [2]	A.2/14 Call changeover
Preamble	idle_to_TX_reservation
Postamble	RX_Reservation_to_idle

DMO_MSMS_DMCC_CM_BV_TR_05	**Reference:** ETS 300 396-3 [1], 6.2.5.1
Purpose	Establish CM call
Test description	The tester issues an implicit send containing a "DMCC_SETUP_request" to the IUT.
Pass criteria	The IUT sends DM-SETUP or DM-SETUP PRES to the tester according to the IUT capability
Selection ETS 300 396-8-1 [2]	A.2/3 Initiate call setup with or without presence check OR A.2/4
Preamble	idle_to_TX_reservation
Postamble	In order to clear the call with presence check, the tester sends a DM-DISCONNECT PDU and waits for the DM-RELEASE PDU. In order to clear the call without presence check, the postamble TX_occupation_to_idle is used.

DMO_MSMS_DMCC_CM_BV_TR_06	**Reference:** ETS 300 396-3 [1], 6.2.5.1
Purpose	Receive incoming CM call
Test description	The tester sends DM-SETUP PRES to the IUT
Pass criteria	The IUT sends DM-CONNECT PDU to the tester, as the setup request was accepted by the IUT
Selection ETS 300 396-8-1 [2]	A.2/6 Accept call setup with presence check
Preamble	idle_to_TX_reservation
Postamble	The tester sends the DM-CONNECT ACK PDU and the class is cleared using the postamble RX_occupation_To_Idle.

DMO_MSMS_DMCC_CM_BV_TR_07	**Reference:** ETS 300 396-3 [1], 6.2.5.1
Purpose	receive and reject pre-emption for a new call
Test description	The tester sends DM-PREEMPT containing an inacceptable priority level to the IUT indicating new call
Pass criteria	The IUT sends DM-REJECT PDU to the tester and remains in the same state.
Selection ETS 300 396-8-1 [2]	Initiate_CM_Call
Preamble	idle_to_TX_reservation
Postamble	TX_reservation_to_idle

DMO_MSMS_DMCC_CM_BV_TR_08	**Reference:** ETS 300 396-3 [1], 6.2.5.1
Purpose	receive and reject changeover
Test description	The tester sends DM-TX REQUEST including an unacceptable priority level to the IUT indicating call continuation
Pass criteria	The IUT sends DM-REJECT PDU to the tester
Selection ETS 300 396-8-1 [2]	A.2/14 Changeover
Preamble	idle_to_TX_reservation
Postamble	TX_Reservation_to_idle

6.1.2.6 IUT is in RX reservation state

DMO_MSMS_DMCC_CM_BV_RR_01	**Reference:** ETS 300 396-3 [1], 6.2.5.2
Purpose	Receive incoming CM call
Test description	The tester sends DM-SETUP PRES to the IUT
Pass criteria	The IUT sends DM-CONNECT to the tester, as the setup request was accepted by the IUT
Selection ETS 300 396-8-1 [2]	A.2/6 Accept call setup with presence check
Preamble	idle_to_RX_reservation
Postamble	RX_occupation_to_idle

DMO_MSMS_DMCC_CM_BV_RR_02	**Reference:** ETS 300 396-3 [1], 6.2.6
Purpose	Initiate pre-emption to establish new CM call
Test description	The tester issues an implicit send containing a "DMCC_SETUP_request" to the IUT.
Pass criteria	The IUT sends DM-PREEMPT to the tester, which is accepted by the tester (DM-PRE ACCEPT sent back by the tester). Then the IUT sends DM-SETUP or DM-SETUP PRES to the tester according to the IUT capability
Selection ETS 300 396-8-1 [2]	A.2/13 Initiate a new call by pre-emption
Preamble	idle_to_RX_reservation
Postamble	In order to clear the call with presence check, the tester sends a DM-DISCONNECT PDU and waits for the DM-RELEASE PDU. In order to clear the call without presence check, the postamble TX_occupation_to_idle is used.

DMO_MSMS_DMCC_CM_BV_RR_03	**Reference:** ETS 300 396-3 [1], 6.2.5.2
Purpose	Initiate changeover to establish ongoing CM call
Test description	The tester issues an implicit send containing a "DMCC_SETUP_request" to the IUT.
Pass criteria	The IUT sends DM-TX REQUEST to the tester, which is accepted by the tester (DM-TX ACCEPT sent back by the tester). Then the IUT sends back DM-SETUP or DM-SETUP PRES to the tester according to the IUT capability
Selection ETS 300 396-8-1 [2]	A.2/14 Call changeover
Preamble	idle_to_RX_reservation
Postamble	In order to clear the call with presence check, the tester sends a DM-DISCONNECT PDU and waits for the DM-RELEASE PDU. In order to clear the call without presence check, the postamble TX_occupation_to_idle is used.

DMO_MSMS_DMCC_CM_BV_RR_04	**Reference:** ETS 300 396-3 [1], 6.2.5.2
Purpose	Handle the reject of a changeover request
Test description	The tester issues an implicit send containing a "DMCC_SETUP_request" to the IUT. The IUT sends DM-TX REQUEST to the tester, which is rejected by the tester (DM-REJECT sent back by the tester).
Pass criteria	The IUT remains in the same state "call active RX reservation". To test it, the tester issues again an implicit send containing a "DMCC_SETUP_request" to the IUT. The IUT sends DM-TX REQUEST to the tester.
Selection ETS 300 396-8-1 [2]	A.2/4 Call changeover
Preamble	idle_to_RX_reservation
Postamble	The tester sends DM-DISCONNECT to return the IUT to idle

6.1.3 MS-MS CM timer tests

6.1.3.1 DT303 Response to DM-SETUP PRES timer

DMO_MSMS_DMCC_CM_TI_01	**Reference:** ETS 300 396-3 [1], 6.2.2.1
Purpose	Time out DT303 for response to DM SET UP PRES.
Test description	The tester sends an implicit send (DMCC_SETUP_request) to the IUT to cause a call setup. Then the IUT sends DM-SETUP PRES to the tester, The tester does not answer within DT303 time.
Pass criteria	After DT303 time out, the IUT sends the DM-RELEASE PDU or the DM-SETUP PRES PDU again to the tester, until DN303 attempts are made
Selection ETS 300 396-8-1 [2]	A.2/4 Initiate call setup with presence check
Preamble	None
Postamble	The tester sends back DM-DISCONNECT to reject the call

6.1.3.2 DT311 Call transaction timer

DMO_MSMS_DMCC_CM_TI_02	**Reference:** ETS 300 396-3 [1], 6.2.4.1
Purpose	Initiate end of transmission after time out of DT311 call transaction timer
Test description	after time out on DT311, the IUT sends DM-TX CEASED PDU and enters state Call Active TX Reservation.
Pass criteria	The DM-TX CEASED PDU is received by the tester.
Selection ETS 300 396-8-1 [2]	Initiate_CM_call
Preamble	idle_to_TX_occupation
Postamble	TX_reservation_to_idle

6.2 DMCC Short data service (SDS)

Test group objective: To test the behaviour of the DMCC SDS entity of the IUT.

Condition: IUT implements Short Data Service and for some TPs, together with CM calls.

6.2.1 MS-MS SDS Capability tests

To test the basic capabilities of the SDS module of the IUT, when operating in unacknowledged service.

DMO_MSMS_DMCC_SDS_CA_01	**Reference:** ETS 300 396-3 [1], 6.3.1.1.1
Purpose	Establish a SDS with unacknowledged service
Test description	The tester issues an implicit send containing a "DMCC_SDS_UNITDATA request" to the IUT which selects the appropriate data types according to the IUT capabilities
Pass criteria	The IUT sends DM-SDS_UDATA to the tester, up to DN314 times
Selection ETS 300 396-8-1 [2]	A.8/1 Send unacknowledged SDS, group or individual address OR A.9/1
Preamble	None
Postamble	None

6.2.2 MS-MS SDS Valid behaviour tests

6.2.2.1 IUT is in idle state, channel is free

DMO_MSMS_DMCC_SDS_BV_ID_01	**Reference:** ETS 300 396-3 [1], 6.3.1.1.2
Purpose	Establish an SDS with acknowledged service
Test description	The tester issues an implicit send containing a "DMCC_SDS_DATA request" to the IUT which selects the appropriate data types according to the IUT capabilities. When the tester receives DM-SDS DATA, it sends back DM-SDS ACK to the IUT
Pass criteria	The IUT comes back to idle, and no new DM-SDS DATA is sent by the IUT within a given time (greater than DT316) meaning the SDS call was successful
Selection ETS 300 396-8-1 [2]	A.9/2 Send acknowledged SDS with or without data in ACK OR A.9/3
Preamble	None
Postamble	None

DMO_MSMS_DMCC_SDS_BV_ID_02	**Reference:** ETS 300 396-3 [1], 6.3.1.1.2
Purpose	Handle the reject of an SDS with acknowledged service
Test description	The tester issues an implicit send containing a "DMCC_SDS_DATA request" to the IUT which selects the appropriate data types according to the IUT capabilities. When the tester receives DM-SDS DATA, it sends back DM-REJECT to the IUT.
Pass criteria	The IUT comes back to idle, and no new DM-SDS DATA is sent by the IUT within a given time (greater than DT316) meaning the SDS call was properly aborted
Selection ETS 300 396-8-1 [2]	A.9/2 Send acknowledged SDS without or with data in ACK OR A.9/3
Preamble	None
Postamble	None

DMO_MSMS_DMCC_SDS_BV_ID_03	**Reference:** ETS 300 396-3 [1], 6.3.2.2
Purpose	Receive an incoming SDS with acknowledged service
Test description	The tester sends DM-SDS DATA containing the appropriate data for the IUT capabilities, to the IUT
Pass criteria	The IUT sends back to the tester DM-SDS ACK containing data or not, according to the IUT capabilities
Selection ETS 300 396-8-1 [2]	A.11/2 Receive acknowledged SDS without or with data in ACK OR A.11/3
Preamble	None
Postamble	None

DMO_MSMS_DMCC_SDS_BV_ID_04	**Reference:** ETS 300 396-3 [1], 6.3.2.2
Purpose	Receive an incoming SDS with acknowledged service and with FCS.
Test description	The tester sends the DM-SDS DATA PDU containing the appropriate data depending on the IUT capabilities and including FCS.
Pass criteria	Verify that the IUT sends the DM-SDS ACK PDU containing or not data.
Selection ETS 300 396-8-1 [2]	A.11/2 OR A.11/3 Receive acknowledged SDS without or with data in ACK
Preamble	None
Postamble	None

DMO_MSMS_DMCC_SDS_BV_ID_05	**Reference:** ETS 300 396-3 [1], 6.3.1.1.2
Purpose	Establish an SDS with acknowledged service using the FCS.
Test description	The tester issues an implicit send to cause the IUT to initiate a SDS. When the tester receives the DM-SDS DATA PDU with FCS, it sends back the DM-SDS ACK PDU.
Pass criteria	Verify that the SDS call was successful, i.e. the IUT does not send any DM-SDS DATA PDU again.
Selection ETS 300 396-8-1 [2]	A.9/2 Send acknowledged SDS without or with data in ACK OR A.9/3
Preamble	None
Postamble	None

6.2.2.2 IUT is in idle state, channel is busy

DMO_MSMS_DMCC_SDS_BV_IB_01	**Reference:** ETS 300 396-3 [1], 6.3.1.2
Purpose	Initiate pre-emption then establish a new SDS with acknowledged service
Test description	The tester issues an implicit send containing a "DMCC_SDS_DATA request" to the IUT which selects the appropriate data types according to the IUT capabilities. As the channel is busy, the IUT sends a DM-PREEMPT to the tester which accepts it by answering DM-PRE_ACCEPT.
Pass criteria	The IUT sends DM-SDS DATA to the tester when Pre-emption is accepted.
Selection ETS 300 396-8-1 [2]	A.12/2 Send short data after pre-emption of a CM call (new call) AND and sends acknowledged SDS. (A.9/2 OR A.9/3)
Preamble	idle_channel_occupation
Postamble	None

DMO_MSMS_DMCC_SDS_BV_IB_02	**Reference:** ETS 300 396-3 [1], 6.3.1.2
Purpose	Initiate pre-emption the establish a new SDS with unacknowledged service.
Test description	The tester in the CALL ACTIVE TX OCCUPATION state with an other MS. The tester issues an implicit send to cause the IUT to initiate a SDS transfer. As the channel is busy, the IUT sends the DM-PREEMPT PDU to the tester which accepts it by answering the DM-PRE ACCEPT PDU.
Pass criteria	Verify that the IUT sends the DM-SDS UDATA PDU.
Selection ETS 300 396-8-1 [2]	A.12/2 Send short data after pre-emption of a CM call (new call) AND and sends unacknowledged SDS. (A.9/1 OR A.8/1)
Preamble	idle_channel_occupation
Postamble	None

DMO_MSMS_DMCC_SDS_BV_IB_03	**Reference:** ETS 300 396-3 [1], 6.3.1.2
Purpose	Handle the reject of pre-emption for acknowledged SDS.
Test description	The tester issues an implicit send containing a "DMCC_SDS_DATA request" to the IUT which selects the appropriate data types according to the IUT capabilities. As the channel is busy, the IUT sends a DM-PREEMPT to the tester which does not accept it and answers DM-REJECT.
Pass criteria	The IUT comes back to idle, and no new DM-SDS DATA is sent by the IUT within a given time (greater than DT316) meaning the SDS call was properly aborted
Selection ETS 300 396-8-1 [2]	A.12/2 Send short data after pre-emption of a CM call (new call) AND and sends acknowledged SDS. (A.9/2 OR A.9/3)
Preamble	idle_channel_occupation
Postamble	None

DMO_MSMS_DMCC_SDS_BV_IB_04	**Reference:** ETS 300 396-3 [1], 6.3.1.2
Purpose	Handle the rejection of pre-emption for SDS with unacknowledged service.
Test description	The tester in the CALL ACTIVE TX OCCUPATION state with an other MS. The tester issues an implicit send to cause the IUT to initiate a SDS transfer. As the channel is busy, the IUT sends the DM-PREEMPT PDU to the tester which rejects by answering the DM-REJECT PDU.
Pass criteria	Verify that the IUT does not send the DM-SDS UDATA PDU within a time greater than DT316, meaning that the SDS call was properly aborted.
Selection ETS 300 396-8-1 [2]	A.12/2 Send short data after pre-emption of a CM call (new call) AND and sends unacknowledged SDS. (A.9/1 OR A.8/1)
Preamble	idle_channel_occupation
Postamble	None

6.2.2.3 IUT is in state TX occupation

No TP are possible from this state because though it is an optional feature, the wording of the specifications, using many times the word "may" does not oblige all implementations to behave as described here. See ETS 300 396-3, 6.3.1.3 as example.

6.2.2.4 IUT is in RX occupation state

DMO_MSMS_DMCC_SDS_BV_RO_01	**Reference:** ETS 300 396-3 [1], 6.3.1.4.1
Purpose	Initiate pre-empt then establish ongoing SDS
Test description	The tester issues an implicit send containing a "DMCC_SDS_DATA request" to the IUT which selects the appropriate data types according to the IUT capabilities. As the channel is busy, the IUT sends a DM-PREEMPT to the tester which accepts it by answering DM-PRE_ACCEPT.
Pass criteria	The IUT sends DM-SDS DATA to the tester when Pre-emption is accepted.
Selection ETS 300 396-8-1 [2]	A.12/4 Send short data after pre-emption of a CM call (ongoing AND call) and sends acknowledged SDS. (A.9/2 OR A.9/3)
Preamble	idle_to_RX_occupation
Postamble	None

DMO_MSMS_DMCC_SDS_BV_RO_02	**Reference:** ETS 300 396-3 [1], 6.3.1.4.1
Purpose	Initiate pre-emption to establish ongoing unacknowledged SDS.
Test description	The tester in the CALL ACTIVE TX OCCUPATION state with an other MS. The tester issues an implicit send to cause the IUT to initiate a SDS transfer. As the channel is busy, the IUT sends the DM-PREEMPT PDU to the tester which accepts it by answering the DM-PRE ACCEPT PDU.
Pass criteria	Verify that the IUT sends the DM-SDS UDATA PDU.
Selection ETS 300 396-8-1 [2]	A.12/4 Send short data after pre-emption of a CM call (ongoing AND call) and sends unacknowledged SDS. (A.9/1 OR A.8/1)
Preamble	idle_to_RX_occupation
Postamble	None

DMO_MSMS_DMCC_SDS_BV_RO_03	**Reference:** ETS 300 396-3 [1], 6.3.1.1.4.1
Purpose	Initiate pre-empt then establish new SDS
Test description	The tester issues an implicit send containing a "DMCC_SDS_DATA request" to the IUT which selects the appropriate data types according to the IUT capabilities. As the channel is busy, the IUT sends a DM-PREEMPT to the tester which accepts it by answering DM-PRE_ACCEPT.
Pass criteria	The IUT sends DM-SDS DATA to the tester when Pre-emption is accepted.
Selection ETS 300 396-8-1 [2]	A.12/2 Send short data after pre-emption of a CM call (new call) AND and sends acknowledged SDS. (A.9/2 OR A.9/3)
Preamble	idle_to_RX_occupation
Postamble	None

DMO_MSMS_DMCC_SDS_BV_RO_04	**Reference:** ETS 300 396-3 [1], 6.3.1.1.4.1
Purpose	Initiate pre-emption to establish new unacknowledged SDS.
Test description	The tester in the CALL ACTIVE TX OCCUPATION state with an other MS. The tester issues an implicit send to cause the IUT to initiate a SDS transfer. As the channel is busy, the IUT sends the DM-PREEMPT PDU to the tester which accepts it by answering the DM-PRE ACCEPT PDU.
Pass criteria	Verify that the IUT sends the DM-SDS UDATA PDU.
Selection ETS 300 396-8-1 [2]	A.12/2 Send short data after pre-emption of a CM call (new call) AND and sends unacknowledged SDS. (A.9/1 OR A.8/1)
Preamble	idle_to_RX_occupation
Postamble	None

DMO_MSMS_DMCC_SDS_BV_RO_05	**Reference:** ETS 300 396-3 [1], 6.3.1.4.1
Purpose	Handle the rejection of pre-emption to establish ongoing acknowledged SDS.
Test description	The tester in the CALL ACTIVE TX OCCUPATION state with an other MS. The tester issues an implicit send to cause the IUT to initiate a SDS transfer. As the channel is busy, the IUT sends the DM-PREEMPT PDU to the tester which rejects it by answering the DM-PRE REJECT PDU.
Pass criteria	Verify that the IUT does not send the DM-SDS DATA PDU.
Selection ETS 300 396-8-1 [2]	A.12/2 Send short data after pre-emption of a CM call (ongoing call) and sends acknowledged SDS. AND (A.9/2 OR A.9/3)
Preamble	idle_to_RX_occupation
Postamble	RX_occupation_to_idle

DMO_MSMS_DMCC_SDS_BV_RO_06	**Reference:** ETS 300 396-3 [1], 6.3.1.4.1
Purpose	Handle the rejection of pre-emption to establish ongoing unacknowledged SDS.
Test description	The tester in the CALL ACTIVE TX OCCUPATION state with an other MS. The tester issues an implicit send to cause the IUT to initiate a SDS transfer. As the channel is busy, the IUT sends the DM-PREEMPT PDU to the tester which rejects it by answering the DM-PRE REJECT PDU.
Pass criteria	Verify that the IUT does not send the DM-SDS UDATA PDU.
Selection ETS 300 396-8-1 [2]	A.12/2 Send short data after pre-emption of a CM call (ongoing call) AND and sends unacknowledged SDS. (A.9/1 OR A.8/1)
Preamble	idle_to_RX_occupation
Postamble	RX_occupation_to_idle

DMO_MSMS_DMCC_SDS_BV_RO_08	**Reference:** ETS 300 396-3 [1], 6.3.1.4.1
Purpose	Handle the rejection of pre-emption to establish new acknowledged SDS.
Test description	The tester in the CALL ACTIVE TX OCCUPATION state with an other MS. The tester issues an implicit send to cause the IUT to initiate a SDS transfer. As the channel is busy, the IUT sends the DM-PREEMPT PDU to the tester which rejects it by answering the DM-PRE REJECT PDU.
Pass criteria	Verify that the IUT does not send the DM-SDS DATA PDU.
Selection ETS 300 396-8-1 [2]	A.12/4 Send short data after pre-emption of a CM call (new call) AND and sends acknowledged SDS. (A.9/2 OR A.9/3)
Preamble	idle_to_RX_occupation
Postamble	RX_occupation_to_idle

DMO_MSMS_DMCC_SDS_BV_RO_09	**Reference:** ETS 300 396-3 [1], 6.3.1.4.1
Purpose	Handle the rejection of pre-emption to establish new unacknowledged SDS.
Test description	The tester in the CALL ACTIVE TX OCCUPATION state with an other MS. The tester issues an implicit send to cause the IUT to initiate a SDS transfer. As the channel is busy, the IUT sends the DM-PREEMPT PDU to the tester which rejects it by answering the DM-PRE REJECT PDU.
Pass criteria	Verify that the IUT does not send the DM-SDS UDATA PDU.
Selection ETS 300 396-8-1 [2]	A.12/2 Send short data after pre-emption of a CM call (new AND call) and sends unacknowledged SDS. (A.9/1 OR A.8/1)
Preamble	idle_to_RX_occupation
Postamble	RX_occupation_to_idle

6.2.2.5 IUT is in TX reservation state

DMO_MSMS_DMCC_SDS_BV_TR_01	**Reference:** ETS 300 396-3 [1], 6.3.1.4.3
Purpose	Initiate SDS from TX_reservation state
Test description	The tester issues an implicit send containing a "DMCC_SDS_DATA request" to the IUT which selects the appropriate data types according to the IUT capabilities. DM-SDS DATA PDU tells it is a transaction within a circuit mode call
Pass criteria	The IUT sends DM-SDS DATA to the tester
Selection ETS 300 396-8-1 [2]	A.12/6 Send SDS as master of a CM call and IUT supports AND acknowledged SDS (A.9/2 OR A.9/3)
Preamble	idle_to_TX_reservation
Postamble	The tester issues a DM-REJECT, followed by TX_Reservation_to_idle

DMO_MSMS_DMCC_SDS_BV_TR_02	**Reference:** ETS 300 396-3 [1], 6.3.1.4.3
Purpose	Initiate unacknowledged SDS from TX reservation state.
Test description	The tester issues an implicit send to cause the IUT to transfer unacknowledged short data.
Pass criteria	Verify that the IUT sends the DM-SDS UDATA PDU
Selection ETS 300 396-8-1 [2]	A.12/6 Send SDS as master of a CM call and IUT supports AND unacknowledge SDS. (A.8/1 OR A.9/1)
Preamble	idle_to_TX_reservation
Postamble	The tester issues a DM-REJECT, followed by TX_Reservation_to_idle

DMO_MSMS_DMCC_SDS_BV_TR_03	**Reference:** ETS 300 396-3 [1], 6.2.5.1
Purpose	Receive incoming acknowledged SDS
Test description	The tester sends DM-SDS DATA to the IUT
Pass criteria	The IUT sends DM-SDS ACK to the tester, meaning the request was accepted by the IUT
Selection ETS 300 396-8-1 [2]	A.11/2 Receive acknowledged SDS without or with data in ACK OR A.11/3
Preamble	idle_to_TX_reservation
Postamble	None

6.2.2.6 IUT is in RX reservation state

DMO_MSMS_DMCC_SDS_BV_RR_01	**Reference:** ETS 300 396-3 [1], 6.2.5.2
Purpose	Receive incoming acknowledged SDS
Test description	The tester sends DM-SDS DATA to the IUT
Pass criteria	The IUT sends DM-SDS ACK to the tester, meaning the request was accepted by the IUT
Selection ETS 300 396-8-1 [2]	A.11/2 Receive acknowledged SDS without or with data in ACK OR A.11/3
Preamble	idle_to_RX_reservation
Postamble	None

DMO_MSMS_DMCC_SDS_BV_RR_02	**Reference:** ETS 300 396-3 [1], 6.2.5.2
Purpose	Receive incoming acknowledged SDS within the CM call..
Test description	The tester sends the DM-SDS DATA PDU to the IUT. The SDS are sent as a transaction within the CM call.
Pass criteria	Verify that the IUT sends back the DM-SDS ACK PDU. Verify that the IUT stay in the RX reservation state.
Selection ETS 300 396-8-1 [2]	A.11/2 Receive acknowledged SDS without or with data in ACK OR A.11/3
Preamble	idle_to_RX_reservation
Postamble	RX_Reservation_to_idle

DMO_MSMS_DMCC_SDS_BV_RR_03	**Reference:** ETS 300 396-3 [1], 6.3.1.4.2
Purpose	Initiate changeover then establish ongoing SDS
Test description	The tester issues an implicit send containing a "DMCC_SDS_DATA request" to the IUT which selects the appropriate data types according to the IUT capabilities. As the channel is busy, the IUT sends a DM-TX REQUEST to the tester which accepts it by answering DM-TX ACCEPT.
Pass criteria	The IUT sends DM-SDS DATA to the tester when changeover is accepted.
Selection ETS 300 396-8-1 [2]	A.12/5 Send acknowledged SDS after changeover AND (A.9/2 OR A.9/3)
Preamble	idle_to_RX_reservation
Postamble	Tester sends the DM-SDS ACK PDU and TX_Reservation_to_idle

DMO_MSMS_DMCC_SDS_BV_RR_04	**Reference:** ETS 300 396-3 [1], 6.3.1.4.2
Purpose	Initiate changeover then establish ongoing unacknowledged SDS
Test description	The tester issues an implicit send containing a "DMCC_SDS_UDATA request" to the IUT which selects the appropriate data types according to the IUT capabilities. As the channel is busy, the IUT sends a DM-TX REQUEST to the tester which accepts it by answering DM-TX ACCEPT.
Pass criteria	The IUT sends DM-SDS UDATA to the tester when changeover is accepted.
Selection ETS 300 396-8-1 [2]	A.12/5 Send unacknowledged SDS after changeover AND (A.9/1 OR A.8/1)
Preamble	idle_to_RX_reservation
Postamble	Tester sends the DM-SDS ACK PDU and TX_Reservation_to_idle

6.2.3 MS-MS SDS Timer tests

6.2.3.1 DT316 Response to DM-SDS DATA timer

DMO_MSMS_DMCC_SDS_TI_01	**Reference:** ETS 300 396-3 [1], 6.3.1.1.2
Purpose	Time out on DT316 timer and retry an SDS DATA with acknowledged service
Test description	The tester issues an implicit send containing a "DMCC_SDS_DATA request" to the IUT which selects the appropriate data types according to the IUT capabilities. When the tester receives DM-SDS DATA, it waits and DOES NOT send back DM-SDS ACK to the IUT within DT316
Pass criteria	The IUT sends a new DM-SDS DATA within a given time (greater than DT316) and for a number of times less than DN316 attempt number, meaning the time out for SDS response was successful
Selection ETS 300 396-8-1 [2]	A.9/2 Send acknowledged SDS without or with data in ACK OR A.9/3
Preamble	None
Postamble	The tester sends back DM-SDS ACK to the IUT

6.3 DMO MS-MS MAC layer

6.3.1 MS-MS MAC capability tests

Test group objective: To test DM-MAC basic capability: fill bit mechanism.

DMO_MSMS_MAC_CA_01	**Reference:** ETS 300 396-3 [1], 8.5.5.1
Purpose	Fill bit addition mechanism in sending mode.
Test description	The tester issues an implicit send to cause the IUT to initiate a CM or SDS call. The IUT sends a DMAC-SYNC containing DM-SETUP or DM-SETUP PRES or DM-SDS DATA or DM-SDS UDATA SDU.
Pass criteria	Check that DMAC-SYNC PDU sent by the IUT is correct, meaning that the IUT fill bit addition mechanism works properly
Selection ETS 300 396-8-1 [2]	Initiate_CM_or_SDS_call
Preamble	None
Postamble	In the case of CM call: 1) terminate to establish the call if CM call with presence check 2) then TX_occupation_to_idle

DMO_MSMS_MAC_CA_02	**Reference:** ETS 300 396-3 [1], 8.5.5.2
Purpose	Fill bit deletion mechanism in sending mode.
Test description	The tester initiates a CM / SDS call by transmitting to the IUT a DMAC-SYNC PDU containing DM-SETUP PRES SDU.
Pass criteria	Check that the IUT sends back the DMAC-SYNC PDU containing the DM-CONNECT SDU, meaning that the IUT fill bit deletion mechanism works properly.
Selection ETS 300 396-8-1 [2]	A.2/6 IUT supports the receipt of call setup with presence check
Preamble	None
Postamble	RX_occupation_to_idle

6.3.2 MS-MS MAC valid behaviour tests

6.3.2.1 DM channel usage procedures

Test group objective: To test DM channel usage procedures of the DM-MAC entity.

DMO_MSMS_MAC_BV_CU_01	**Reference:** ETS 300 396-3 [1], 8.4.5.1
Purpose	Initiation of CM or SDS call in DSB.
Test description	The tester issues an implicit send to cause the IUT to initiate a CM or SDS call.
Pass criteria	Verify that the IUT sends the DM-SETUP or DM-SETUP PRES or DM-SDS DATA or DM-SDS UDATA SDU in DSB.
Selection ETS 300 396-8-1 [2]	Initiate_CM_or_SDS_call
Preamble	None
Postamble	In the case of CM call: 1) terminate to establish the call if CM call with presence check 2) then TX_occupation_to_idle

DMO_MSMS_MAC_BV_CU_02	**Reference:** ETS 300 396-3 [1], 8.5.1, 8.4.5.1.7
Purpose	Transmission of the DM-OCCUPIED SDU when the channel is busy.
Test description	The tester sends an implicit send to cause the IUT to initiate a CM call with or without presence check.
Pass criteria	Verify that once the channel is occupied, the IUT generates the DM-OCCUPIED SDU in time slot 3 of frames 6, 12 and 18.
Selection ETS 300 396-8-1 [2]	Initiate_CM_call
Preamble	Idle_to_TX_occupation
Postamble	TX_occupation_to_idle

DMO_MSMS_MAC_BV_CU_03	**Reference:** ETS 300 396-3 [1], 8.4.6.1
Purpose	Generation and transmission of layer 2 DM-RESERVED SDU.
Test description	The IUT MAC starts transmitting the DM-RESERVED SDUs.
Pass criteria	Check that DMAC-SYNC containing DM-RESERVED SDUs are sent in timeslots 1 and 3 of frames 6, 12, and 18 using the same priority level as for the DM-TX CEASED SDUs.
Selection ETS 300 396-8-1 [2]	Initiate_CM_call
Preamble	idle_to_TX_occupation
Postamble	TX occupation to idle

DMO_MSMS_MAC_BV_CU_04	**Reference:** ETS 300 396-3 [1], 8.4.6.1
Purpose	The sending of the DM-RESERVED SDU stopped when the reservation period expired.
Test description	The tester issues an implicit send to cause the IUT to send the DM-TX CEASED SDU.
Pass criteria	Verify that in CALL ACTIVE TX RESERVATION STATE, the IUT sends the DMAC-SYNC PDU containing the DM-RESERVED until the "reservation time remaining" equals 0.
Selection ETS 300 396-8-1 [2]	Initiate_CM_call
Preamble	idle_to_TX_occupation
Postamble	None

DMO_MSMS_MAC_BV_CU_05	**Reference:** ETS 300 396-3 [1], 8.4.6.2
Purpose	Transmission of DM-SDS OCCUPIED SDU when transmitting SDS data.
Test description	The tester issues an implicit to cause the IUT to initiate a SDS call. Then the IUT sends the DMAC-SYNC PDU containing the DM-SDS DATA or DM-SDS UDATA SDU.
Pass criteria	During the transmission of the SDS data, the IUT issues DMAC-SYNC containing DM-SDS OCCUPIED SDU. It is transmitted in DSB in time slot 3 of frames 6 and 12 and in time slots 1 and 3 of frame 18.
Selection ETS 300 396-8-1 [2]	A.6/1 Short Data Service send data
Preamble	None
Postamble	None

DMO_MSMS_MAC_BV_CU_06	**Reference:** ETS 300 396-3 [1], 8.4.7.1, 8.4.7.2, 8.5.6.1
Purpose	Specified number of re-transmission is fulfilled with respect to the frame count down element.
Test description	The tester issues an implicit send to cause the IUT to initiate a CM or SDS call. The IUT is transmitting a DMAC-SYNC PDU containing DM-SETUP or DM-SETUP PRES or DM-SDS DATA or DM-SDS UDATA SDU, repeated in the number of frames indicated by the frame count down element.
Pass criteria	The number of repeated transmissions in consecutive frames corresponds to the value provided in the frame count down element, and the PDU is not repeated after the one with frame count down element value 0 (absence observed during a period of time).
Selection ETS 300 396-8-1 [2]	Initiate_CM_or_SDS_call
Preamble	None
Postamble	None

DMO_MSMS_MAC_BV_CU_07	**Reference:** ETS 300 396-3 [1], 8.4.7.5, 8.5.4
Purpose	Fragmentation.
Test description	The tester issues an implicit send such that the IUT initiates a SDS by transmitting DM-SDS DATA or DM-SDS UDATA PDU with data type 2, 3 or 4 in order to receive a fragmented message.
Pass criteria	DMAC-SYNC with Fragmentation flag set to value 1, followed by n times DMAC-FRAG then ending with DMAC-END
Selection ETS 300 396-8-1 [2]	A.38/5 Fragmentation and user defined data 2,3 or 4 and one of AND the conditions expressed in: Initiate_SDS_call (A.13/4 OR A.13/3 OR A.13/2) AND Initiate_SDS_call
Preamble	None
Postamble	None

DMO_MSMS_MAC_BV_CU_08	**Reference:** ETS 300 396-3 [1], 8.4.7.12
Purpose	Channel A usage.
Test description	The tester issues an implicit send such that the IUT initiates a CM or SDS call. The IUT sends a DMAC-SYNC containing a DM-SETUP or DM-SETUP PRES or DM-SDS DATA or DM-SDS UDATA PDU according to the IUT capabilities
Pass criteria	The A/B channel usage in DMAC-SYNC is set to value 00 or 01, meaning A channel usage.
Selection ETS 300 396-8-1 [2]	Initiate_CM_or_SDS_call
Preamble	None
Postamble	None

6.3.2.2 Signalling messages procedures

Test group objective: To test the signalling procedures of the DM-MAC entity.

DMO_MSMS_MAC_BV_SM_01	**Reference:** ETS 300 396-3 [1], 8.5.2.1.1
Purpose	Addressing in synchronization burst for initiation of a group addressed call.
Test description	The tester issues an implicit send to cause the IUT to initiate a CM or SDS call. The IUT sends a DMAC-SYNC PDU containing a DM-SETUP or DM-SETUP PRES or DM-SDS DATA or DM-SDS UDATA SDU.
Pass criteria	Verify the SSI and MNI destination elements in the DMAC-SYNC header and verify that the destination address type is set to 0.
Selection ETS 300 396-8-1 [2]	A.38/1 Addressing in synchronization burst and AND one of the conditions expressed in Initiate_CM_or_SDS_call Initiate_CM_or_SDS_call
Preamble	None
Postamble	None

DMO_MSMS_MAC_BV_SM_02	**Reference:** ETS 300 396-3 [1], 8.5.7.3.2
Purpose	Synchronization burst for a random access message.
Test description	The tester issues an implicit send to cause the IUT to initiate pre-emption. As the channel is busy, the IUT sends a DM-PREEMPT request (address = master) to the tester.
Pass criteria	Check that the DM-PREEMPT request is sent using DMAC-SYNC PDU
Selection ETS 300 396-8-1 [2]	A.2/10 Initiate pre-emption in ongoing call
Preamble	idle_to_RX_occupation
Postamble	Tester issues a DM-REJECT followed by RX_occupation_to_idle

DMO_MSMS_MAC_BV_SM_03	**Reference:** ETS 300 396-3 [1], 8.5.2.1.1
Purpose	Addressing in synchronization burst for a random access message.
Test description	The tester issues an implicit to cause the IUT to initiate a CM call. The IUT sends DM-PREEMPT (address = master) to the tester.
Pass criteria	The destination address of the DMAC-SYNC containing DM-PREEMPT sent by the IUT is the current master DM-MS layer 2 address.
Selection ETS 300 396-8-1 [2]	A.38/1 Addressing in synchronization burst and Initiate pre-emption in ongoing call AND and one of the conditions expressed in: A.2/12 Initiate_CM_call AND Initiate_CM_call
Preamble	idle_to_RX_occupation
Postamble	Tester issues a DM-REJECT followed by RX_occupation_to_idle

DMO_MSMS_MAC_BV_SM_04	**Reference:** ETS 300 396-3 [1], 8.5.2.1.1
Purpose	Addressing in synchronization burst in the DM-OCCUPIED PDU.
Test description	The tester sends an implicit send to cause the IUT to initiate a CM call. The IUT sends the DMAC-SYNC PDU containing the DM-SETUP or DM-SETUP PRES SDU. Once the call is established (the channel is busy), the IUT sends the DMAC-SYNC PDU containing the DM-OCCUPIED SDU.
Pass criteria	The MNI and source address elements in a DMAC-SYNC containing DM-OCCUPIED SDU are the same as the ones used in the DM-SETUP.
Selection ETS 300 396-8-1 [2]	A.38/1 Addressing in synchronization burst AND and one of the conditions expressed in: Initiate_CM_call Initiate_CM_call
Preamble	None
Postamble	TX_occupation_to_idle

DMO_MSMS_MAC_BV_SM_05	**Reference:** ETS 300 396-3 [1], 8.4.7.5, 8.5.4.1
Purpose	Fragmentation PDUs are sent in consecutive frames.
Test description	The tester issues an implicit send to cause the IUT to initiate a SDS call with fragmentation.
Pass criteria	Verify that the DMAC-FRAG PDUs and DMAC-END PDU are sent in consecutive frames.
Selection ETS 300 396-8-1 [2]:	A.38/5 AND Fragmentation and (A.13/2 OR User defined data 4 or 2 or 3 and A.13/3 OR A.13/4) AND (A.9/2 OR A.9/3) Send acknowledged SDS with or without data in ACK
Preamble	None
Postamble	None

DMO_MSMS_MAC_BV_SM_06	**Reference:** ETS 300 396-3 [1], 8.4.7.5, 8.5.4.1
Purpose	For acknowledged data message sent using fragmentation, if the acknowledge is sent to the IUT then no re-transmission takes place.
Test description	The tester issues an implicit send to cause the IUT to initiate a SDS call with fragmentation. The IUT sends the DMAC-SYNC, DMAC FRAG and DMAC END PDUs.
Pass criteria	Verify that after receipt of the acknowledge SDU, the IUT does not re-transmit the SDS data..
Selection ETS 300 396-8-1 [2]:	A.38/5 AND Fragmentation and (A.13/2 OR User defined data 4 or 2 or 3 and A.13/3 OR A.13/4) AND (A.9/2 OR A.9/3) Send acknowledged SDS with or without data in ACK
Preamble	None
Postamble	None

DMO_MSMS_MAC_BV_SM_07	**Reference:** ETS 300 396-3 [1], 8.5.4.2	
Purpose	Reconstruction procedure for acknowledged SDS data messages.	
Test description	The tester sends a fragmented SDS data type 2 3 or 4 message.	
Pass criteria	Check that the IUT sends back a DMAC-SYNC containing SDS-DATA ACK, indicating that the message was received without error.	
Selection ETS 300 396-8-1 [2]	A.38/6 AND A.13/2 AND A.13/3 AND A.13/4 AND (A.11/2 OR A.11/3)	Reconstruction and User defined data 2 and User defined data 3 and User defined data 4 and Receive acknowledged SDS with or without data in ACK
Preamble	None	
Postamble	None	

DMO_MSMS_MAC_BV_SM_08	**Reference:** ETS 300 396-3 [1], 8.5.7.3.6
Purpose	Abandoning random access attempt. (DN213)
Test description	The tester issues an implicit send to cause the IUT to initiate pre-emption. The IUT sends DM-PREEMPT request (address = master) to the tester. The tester does not answer the request by DM-PRE ACCEPT
Pass criteria	The IUT stops sending DMAC-SYNC containing DM-PREEMPT after DN213 times for a non emergency message and 2*DN213 for an emergency message.
Selection ETS 300 396-8-1 [2]	A.2/12 Initiate pre-emption in ongoing call
Preamble	idle_to_RX_occupation
Postamble	None

DMO_MSMS_MAC_BV_SM_09	**Reference:** ETS 300 396-3 [1], 8.5.7.2.1
Purpose	Pre-emption flag in the DM-OCCUPIED SDU.
Test description	In TX occupation state, the IUT generates and sends the DMAC-SYNC PDU containing the DM-OCCUPIED SDU.
Pass criteria	Verify that when generating the DM-OCCUPIED SDU, the IUT set the pre-emption request flag to 1.
Selection ETS 300 396-8-1 [2]	Initiate_CM_call
Preamble	idle_to_TX_occupation
Postamble	TX_occupation_to_idle

DMO_MSMS_MAC_BV_SM_10	**Reference:** ETS 300 396-3 [1], 8.5.7.2.1
Purpose	Pre-emption flag in the DM-RESERVED SDU.
Test description	In TX reservation state, the IUT generates and sends the DMAC-SYNC PDU containing the DM-RESERVED SDU.
Pass criteria	Verify that when generating the DM-RESERVED SDU, the IUT set the requests flag and the changeover flag to 1.
Selection ETS 300 396-8-1 [2]	Initiate_CM_call
Preamble	idle_to_TX_Reservation
Postamble	TX_Reservation_to_idle

DMO_MSMS_MAC_BV_SM_11	**Reference:** ETS 300 396-3 [1], subclause 8.5.7.3.6
Purpose	Cease random access attempt for timing request after receipt of a rejection.
Test description	The tester issues an implicit send to cause the IUT to initiate a timing change request. The IUT sends the DMAC-SYNC PDU containing the DM-TIMING REQUEST SDU, to the tester that answers the DMAC-SYNC PDU containing the DM-TIMING ACK SDU with a reject.
Pass criteria	Verify that the IUT accept this rejection and does not send the timing change request any more.
Selection ETS 300 396-8-1 [2]	IUT accepts CM call.
Preamble	idle_to_RX_Occupation
Postamble	RX_Occupation_to_idle

6.3.4 MS-MS MAC timer tests

DMO_MSMS_MAC_BV_TI_01	**Reference:** ETS 300 396-3 [1], 8.5.7.2.3
Purpose	Response to a pre-emption request within time DT211.
Test description	The tester sends a DM-PREEMPT to the IUT, containing the address of the master. The IUT sends back DM-PRE ACCEPT
Pass criteria	check that the IUT MAC sends back DMAC-SYNC containing DM-PRE ACCEPT within time DT211, and that it repeats the same DM-PRE ACCEPT SDU the number of frames specified
Selection ETS 300 396-8-1 [2]	A.2/11 Accept call pre-emption
Preamble	idle_to_TX_occupation
Postamble	RX_Reservation_to_idle

Annex A (informative): Bibliography

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